

$$\Rightarrow 4x = -(x^2 + y^2 - 4) \Rightarrow x^2 + 4x + y^2 = 4$$

Completing the square for  $x^2$ , we have

$$(x + 2)^2 + y^2 = 8$$

### 1.6 Complex Numbers in the Real World

**Example 1:** If the impedance of circuit is  $11(\cos 55.35^\circ + i \sin 55.35^\circ)$  ohms at a voltage of  $25(\cos 30^\circ + i \sin 30^\circ)$  V, find the value of current in the circuit.

**Sol.** Substitute the voltage  $25(\cos 30^\circ + i \sin 30^\circ)$  and impedance  $11(\cos 55.35^\circ + i \sin 55.35^\circ)$  into the equation  $V = IZ$ , where  $V$  is voltage,  $I$  denote the current and  $Z$  is impedance.

$$25(\cos 30^\circ + i \sin 30^\circ) = I \cdot 11(\cos 55.35^\circ + i \sin 55.35^\circ)$$

$$\text{or } I = \frac{25(\cos 30^\circ + i \sin 30^\circ)}{11(\cos 55.35^\circ + i \sin 55.35^\circ)}$$

$$I = \frac{25}{11} [\cos(30^\circ - 55.35^\circ) + i \sin(30^\circ - 55.35^\circ)]$$

$$I = 2.27 [\cos(-25.35^\circ) + i \sin(-25.35^\circ)]$$

Express into rectangular form

$$I = 2.27 [0.90 + i(-0.42)] = 2.04 - 0.95i$$

Thus, current is  $2.04 - 0.95i$

**Note:**

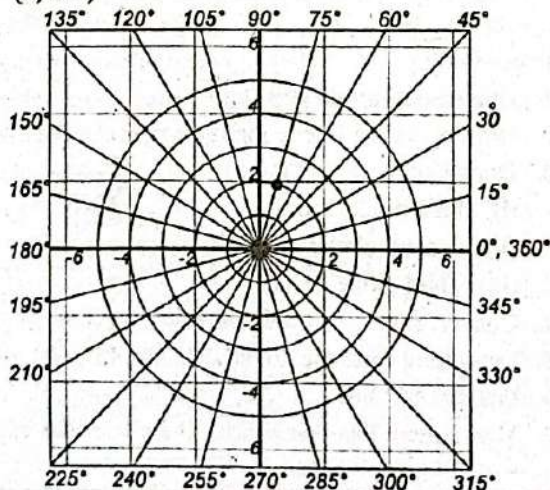
1.  $(-5, \pi/4)$  and  $(5, 5\pi/4)$  represent the same point in the plane.
2. We can write  $\cos \theta + i \sin \theta = \text{cis } \theta$
3. (i) If  $x = 0, y > 0$  then  $\theta = 90^\circ$   
(ii) If  $x = 0, y < 0$  then  $\theta = -90^\circ$   
(iii) If  $x = 0, y = 0$  then  $\theta$  is undefined.

### Exercise 1.5

**Q1:** Plot the following points:

(i)  $(2, 75^\circ)$

**Sol.**

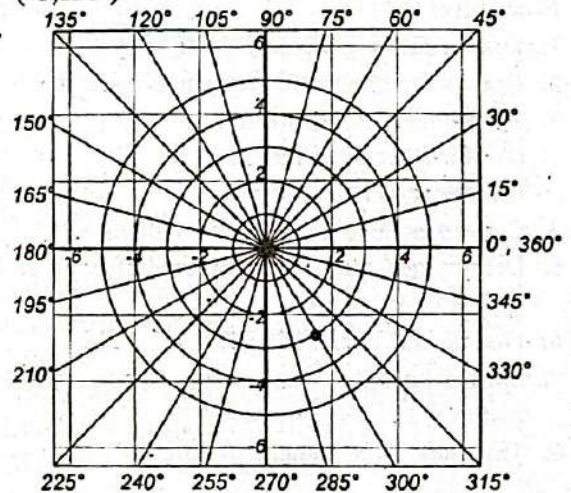


### Procedure:

1. Start with a dot at the origin.
2. Draw a horizontal line to the right from the dot.
3. Make circles around the dot:  
(i) First circle with radius 1.  
(ii) Second circle with radius 2.
4. Draw lines from the dot at different angles ( $15^\circ, 30^\circ, 45^\circ$ , etc.)
5. Find the line at  $75^\circ$  from the horizontal.
6. On this  $75^\circ$  line, mark where it crosses the second circle (radius 2).
7. This marked spot is the point  $(2, 75^\circ)$ .

(ii)  $(-3, 120^\circ)$

**Sol.**



### Procedure:

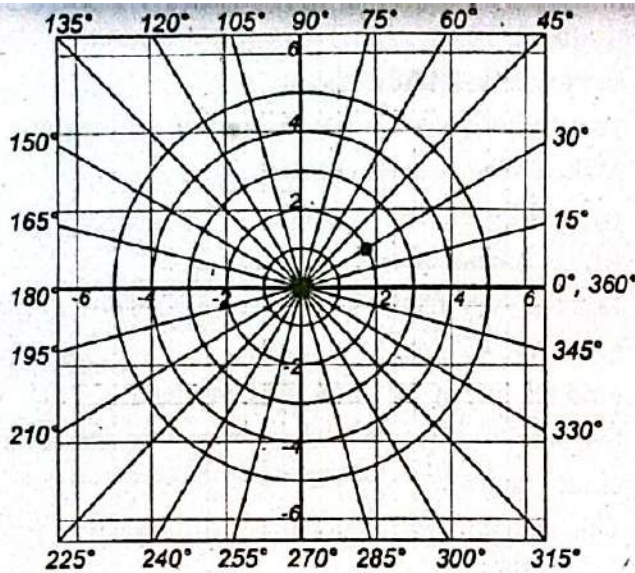
1. Draw a dot (origin point).
2. Draw a straight line to the right from the dot.
3. Draw circles around the dot:  
(i) First circle: radius = 1  
(ii) Second circle: radius = 2  
(iii) Third circle: radius = 3
4. Draw lines from the dot at different angles ( $15^\circ, 30^\circ, 45^\circ, 60^\circ$ , etc.).
5. Find the  $120^\circ$  line.
6. Since radius negative  $(-3)$ , go to the opposite side ( $120^\circ + 180^\circ = 300^\circ$ ).
7. Mark where the  $300^\circ$  line crosses the third circle (radius = 3).
8. This mark is the point  $(-3, 120^\circ)$ .

(iii)  $(2, \frac{\pi}{6})$

**Sol.** First we convert radian into degree

$$\frac{\pi}{6} \text{ radian} = \frac{\pi}{6} \times \frac{180}{\pi} = 30^\circ$$





#### Procedure:

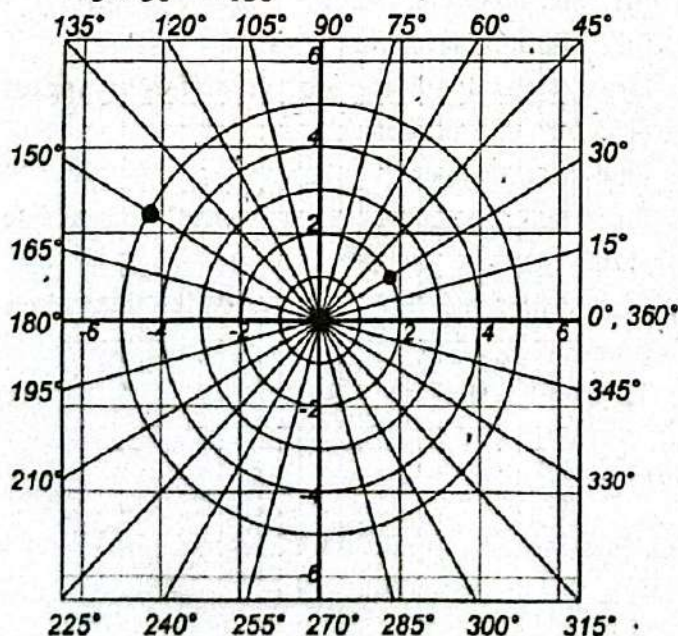
1. Draw a dot (origin point).
2. Draw a straight line to the right from the dot.
3. Draw circles around the dot:
  - (i) First circle: radius = 1
  - (ii) Second circle: radius = 2
4. Convert angle to degrees:  $\pi/6$  radians =  $30^\circ$
5. Draw lines from the dot at different angles ( $15^\circ, 30^\circ, 45^\circ$ , etc.).
6. Find the  $30^\circ$  line.
7. Mark where this line crosses the second circle (radius = 2).
8. This mark is the point  $(2, \pi/6)$ .

(iv)  $\left(5, \frac{5\pi}{6}\right)$

Sol. First, we convert radian into degree

$$\frac{5\pi}{6} \text{ radian} = \frac{5\pi}{6} \times \frac{180^\circ}{\pi} = \frac{5}{6} \times 180^\circ$$

$$= 5 \times 30^\circ = 150^\circ$$



#### Procedure:

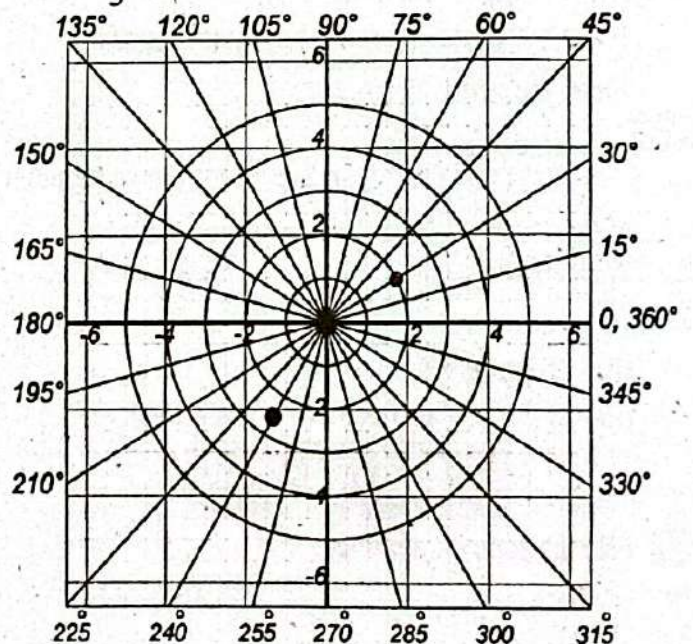
1. Draw a dot (origin point)
2. Draw a straight line to the right from the dot.
3. Draw circles around the dot:
  - (i) First circle: radius = 1
  - (ii) Second circle: radius = 2
4. Convert angle:  $5\pi/6$  radians =  $150^\circ$
5. Draw lines from the dot at  $15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ$ , etc.
6. Find the  $150^\circ$  line
7. Mark where this line crosses the fifth circle (radius = 5)
8. This mark is the point  $(5, 5\pi/6)$ .

(v)  $\left(-\frac{5}{2}, \frac{\pi}{3}\right)$

Sol. First we convert radian into degree.

$$\frac{\pi}{3} \text{ radian} = \frac{\pi}{3} \times \frac{180}{\pi}$$

$$= \frac{180}{3} = 60^\circ$$



#### Procedure:

1. Draw a dot (origin point)
2. Draw a straight line to the right from the dot
3. Draw circles around the dot:
  - (i) First circle: radius = 1
  - (ii) Second circle: radius = 2
  - (iii) Third circle: radius = 3
4. Convert angle:  $\pi/3$  radians =  $60^\circ$
5. Draw lines from the dot at  $15^\circ, 30^\circ, 45^\circ, 60^\circ$ , etc.
6. Find the  $60^\circ$  line
7. Mark where this line crosses the circle with radius = 2.5



**Procedure:**

1. Draw a dot (origin point)
2. Draw a straight line to the right from the dot
3. Draw circles around the dot:
  - (i) First circle: radius = 1
  - (ii) Second circle: radius = 2
  - (iii) Third circle: radius = 3
4. Convert angle:  $5\pi/12$  radians =  $75^\circ$
5. Draw lines from the dot at  $15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ$ , etc.
6. Find the  $75^\circ$  line
7. Mark where this line crosses the circle with radius = 2.5
8. This mark is the point  $(5/2, 5\pi/12)$

(Note: Negative radius means plot in opposite direction, so use  $75^\circ + 180^\circ = 255^\circ$  line instead)

**02. Express the following complex numbers in polar form:**

(i)  $4 + 3i$

Sol. Here,  $x = 4, y = 3$

By using formula

$$r = \sqrt{x^2 + y^2}$$

$$r = \sqrt{4^2 + 3^2}$$

$$r = \sqrt{16 + 9}$$

$$r = \sqrt{25} \Rightarrow r = 5$$

$$\text{Also, } \theta = \tan^{-1}\left(\frac{y}{x}\right) \quad \because z \text{ lies in quadrant I}$$

$$\theta = \tan^{-1}\left(\frac{3}{4}\right) \Rightarrow \theta = 36.87^\circ$$

Now,

$$x + yi = r(\cos \theta + i \sin \theta)$$

$$4 + 3i = 5(\cos 36.87^\circ + i \sin 36.87^\circ)$$

(ii)  $1 + i$

Sol. Here,  $x = 1, y = 1$

By using formula

$$r = \sqrt{x^2 + y^2}$$

$$r = \sqrt{1^2 + 1^2}$$

$$r = \sqrt{1 + 1}$$

$$r = \sqrt{2}$$

Also,

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) \quad \because z \text{ lies in quadrant I}$$

$$\theta = \tan^{-1}\left(\frac{1}{1}\right) \Rightarrow \theta = \frac{\pi}{4}$$

Now,

$$x + yi = r(\cos \theta + i \sin \theta)$$

$$1 + i = \sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$$

$$(iii) \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$\text{Sol. Here, } x = \frac{1}{2}, y = \frac{\sqrt{3}}{2}$$

By using formula

$$r = \sqrt{x^2 + y^2}$$

$$r = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$r = \sqrt{\frac{1}{4} + \frac{3}{4}}$$

$$r = \sqrt{\frac{1+3}{4}}$$

$$r = \sqrt{\frac{4}{4}}$$

$$r = \sqrt{1}$$

$$r = 1$$

$$\text{Also, } \theta = \tan^{-1}\left(\frac{y}{x}\right) \quad \because z \text{ lies in quadrant I}$$

$$\theta = \tan^{-1}\left(\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}\right)$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right)$$

$$\theta = \tan^{-1}(\sqrt{3})$$

$$\theta = \frac{\pi}{3}$$

$$\text{Now, } x + yi = r(\cos \theta + i \sin \theta)$$

$$\frac{1}{2} + \frac{\sqrt{3}}{2}i = 1\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$$

$$(iv) -\frac{5}{2} - \frac{5\sqrt{3}}{2}i$$

$$\text{Sol. Here, } x = -\frac{5}{2}, y = -\frac{5\sqrt{3}}{2}$$

By using formula

$$r = \sqrt{x^2 + y^2}$$

$$r = \sqrt{\left(-\frac{5}{2}\right)^2 + \left(-\frac{5\sqrt{3}}{2}\right)^2}$$



$$r = \sqrt{\frac{25}{4} + \frac{25(3)}{4}}$$

$$r = \sqrt{\frac{25}{4} + \frac{75}{4}}$$

$$r = \sqrt{\frac{25+75}{4}}$$

$$r = \sqrt{\frac{100}{4}}$$

$$r = \sqrt{25} \Rightarrow r = 5$$

$$\text{Also, } \theta = \pi + \tan^{-1}\left(\frac{|y|}{|x|}\right) \because z \text{ lies quadrant III}$$

$$\theta = \pi + \tan^{-1}\left(\frac{\left|\frac{5\sqrt{3}}{2}\right|}{\left|\frac{5}{2}\right|}\right)$$

$$\theta = \pi + \tan^{-1}\left(\frac{\frac{5\sqrt{3}}{2}}{\frac{5}{2}}\right)$$

$$\theta = \pi + \tan^{-1}\left(\frac{\sqrt{3}}{1}\right)$$

$$\theta = \pi + \tan^{-1}(\sqrt{3})$$

$$\theta = \pi + \frac{\pi}{3}$$

$$\theta = \frac{3\pi + \pi}{3}$$

$$\theta = \frac{4\pi}{3}$$

$$\text{Now, } x + yi = r(\cos \theta + i \sin \theta)$$

$$-\frac{5}{2} - \frac{5\sqrt{3}}{2}i = 5\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$$

$$(v) \frac{1-i}{1+i}$$

$$\begin{aligned} \text{Sol. } \frac{1-i}{1+i} &= \frac{1-i}{1+i} \times \frac{1-i}{1-i} \\ &= \frac{1-i-i+i^2}{1^2-i^2} = \frac{1-i-i+(-1)}{1-(-1)} \\ &= \frac{1-i-i-1}{1+1} = \frac{-2i}{2} \\ &= -i = 0-i \end{aligned}$$

$$\text{Here, } x = 0, y = -1$$

By using formula

$$r = \sqrt{x^2 + y^2}$$

$$r = \sqrt{(0)^2 + (-1)^2}$$

$$r = \sqrt{0+1}$$

$$r = \sqrt{1}$$

$$r = 1$$

$$\text{Also, } \theta = \frac{3\pi}{2} \because z \text{ lies on negative } y\text{-axis}$$

Now,

$$x + yi = r(\cos \theta + i \sin \theta)$$

$$0 - i = 1\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$$

$$\frac{1-i}{1+i} = \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}$$

$$(vi) \frac{\sqrt{3}+i}{1+\sqrt{3}i}$$

$$\text{Sol. } \frac{\sqrt{3}+i}{1+\sqrt{3}i} = \frac{\sqrt{3}+i}{1+\sqrt{3}i} \times \frac{1-\sqrt{3}i}{1-\sqrt{3}i}$$

$$= \frac{\sqrt{3} - (\sqrt{3})^2 i + i - \sqrt{3}i^2}{1^2 - (\sqrt{3}i)^2}$$

$$= \frac{\sqrt{3} - 3i + i + \sqrt{3}}{1+3} = \frac{2\sqrt{3} - 2i}{4}$$

$$= \frac{2(\sqrt{3}-i)}{4} = \frac{\sqrt{3}-i}{2} = \frac{\sqrt{3}}{2} - \frac{1}{2}i$$

$$\text{Here, } x = \frac{\sqrt{3}}{2}, y = -\frac{1}{2}$$

By using formula

$$r = \sqrt{x^2 + y^2}$$

$$r = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(-\frac{1}{2}\right)^2}$$

$$r = \sqrt{\frac{3}{4} + \frac{1}{4}}$$

$$r = \sqrt{\frac{3+1}{4}}$$

$$r = \sqrt{\frac{4}{4}}$$

$$r = \sqrt{1}$$

$$r = 1$$



Also  $\theta = 2\pi - \tan^{-1}\left(\frac{|y|}{x}\right) \quad \because z \text{ lies in IV Quadrant}$

$$\theta = 2\pi - \tan^{-1}\left(\frac{\left|\frac{-1}{2}\right|}{\frac{\sqrt{3}}{2}}\right)$$

$$\theta = 2\pi - \tan^{-1}\left(\frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)$$

$$\theta = 2\pi - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\theta = 2\pi - \frac{\pi}{6}$$

$$\theta = \frac{12\pi - \pi}{6}$$

$$\theta = \frac{11\pi}{6}$$

Now,

$$x + yi = r(\cos\theta + i\sin\theta)$$

$$\frac{\sqrt{3}}{2} - \frac{1}{2}i = 1\left(\cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6}\right)$$

$$\frac{\sqrt{3} + i}{1 + \sqrt{3}i} = \cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6}$$

(vii)  $\frac{3+4i}{4+3i}$

Sol.  $\frac{3+4i}{4+3i} = \frac{3+4i}{4+3i} \times \frac{4-3i}{4-3i}$   

$$= \frac{12-9i+16i-12i^2}{4^2-(3i)^2} = \frac{12-9i+16i+12}{16+9}$$

$$= \frac{24+7i}{25} = \frac{24}{25} + \frac{7}{25}i$$

Here,  $x = \frac{24}{25}$ ,  $y = \frac{7}{25}$

By using formula

$$r = \sqrt{x^2 + y^2}$$

$$r = \sqrt{\left(\frac{24}{25}\right)^2 + \left(\frac{7}{25}\right)^2}$$

$$r = \sqrt{\frac{576}{625} + \frac{49}{625}}$$

$$r = \sqrt{\frac{576+49}{625}}$$

$$r = \sqrt{\frac{625}{625}}$$

$$r = \sqrt{1}$$

$$r = 1$$

Also,  $\theta = \tan^{-1}\left(\frac{y}{x}\right) \quad \because z \text{ lies in I Quadrant}$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\theta = \tan^{-1}\left(\frac{\frac{7}{25}}{\frac{24}{25}}\right)$$

$$\theta = \tan^{-1}\left(\frac{7}{24}\right)$$

$$\theta = 16.26^\circ$$

Now,

$$x + yi = r(\cos\theta + i\sin\theta)$$

$$\frac{24}{25} + \frac{7}{25}i = 1(\cos 16.26^\circ + i\sin 16.26^\circ)$$

$$\frac{3+4i}{4+3i} = \cos 16.26^\circ + i\sin 16.26^\circ$$

**Q3.** Convert each of the complex number  $z$  in the rectangular for  $x + iy$ :

(i)  $4\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right)$

Sol.  $4\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right) = 4\left[\left(\frac{1}{2}\right) + i\left(\frac{-\sqrt{3}}{2}\right)\right]$

$$4\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right) = 4\left[\frac{1}{2} - \frac{\sqrt{3}}{2}i\right]$$

$$4\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right) = \frac{4}{2} - \frac{4\sqrt{3}}{2}i$$

$$4\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right) = 2 - 2\sqrt{3}i$$

(ii)  $\frac{3}{2}\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right)$

Sol.  $\frac{3}{2}\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right) = \frac{3}{2}\left[\left(\frac{-\sqrt{3}}{2}\right) + i\left(\frac{-1}{2}\right)\right]$

$$\frac{3}{2}\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right) = \frac{3}{2}\left[-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right]$$

$$\frac{3}{2}\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right) = -\frac{3\sqrt{3}}{4} - \frac{3}{4}i$$



(iii)  $|z| = 7, \arg(z) = \frac{23\pi}{12}$

Sol. By using the formula

$$x + yi = r(\cos \theta + i \sin \theta)$$

$$x + yi = 7 \left( \cos \frac{23\pi}{12} + i \sin \frac{23\pi}{12} \right)$$

$$x + yi = 7[(0.97) + i(-0.26)]$$

$$x + yi = 7(0.97 - 0.26i)$$

$$x + yi = 6.76 - 1.81i$$

(iv)  $|z| = 11, \arg(z) = -\frac{11\pi}{12}$

Sol. By using the formula

$$x + yi = r(\cos \theta + i \sin \theta)$$

$$x + yi = 11 \left[ \cos \left( -\frac{11\pi}{12} \right) + i \sin \left( -\frac{11\pi}{12} \right) \right]$$

$$x + yi = 11 \left( \cos \frac{11\pi}{12} - i \sin \frac{11\pi}{12} \right)$$

$$x + yi = 11[(-0.97) - i(0.26)]$$

$$x + yi = 11[-0.97 - 0.26i]$$

$$x + yi = -10.63 - 2.85i$$

(v)  $|z| = \frac{10}{3}, \arg(z) = -\frac{17\pi}{12}$

Sol. By using the formula

$$x + yi = r(\cos \theta + i \sin \theta)$$

$$x + yi = \frac{10}{3} \left[ \cos \left( -\frac{17\pi}{12} \right) + i \sin \left( -\frac{17\pi}{12} \right) \right]$$

$$x + yi = \frac{10}{3} \left( \cos \frac{17\pi}{12} - i \sin \frac{17\pi}{12} \right)$$

$$x + yi = \frac{10}{3} [(-0.26) - i(-0.97)]$$

$$x + yi = \frac{10}{3} [-0.26 + 0.97i]$$

$$x + yi = -0.86 + 3.22i$$

(vi)  $2\cos(-33) + i2\sin(-33)$

Sol.  $2\cos(-33) + i2\sin(-33) = 2\cos 33 - i2\sin 33$

$$2\cos(-33) + i2\sin(-33) = 2(0.84) - i2(0.54)$$

$$2\cos(-33) + i2\sin(-33) = 1.68 - 1.09i$$

Q4. If  $z_1 = 9 \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right)$  and

$$z_2 = 5 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \text{ then find}$$

(i)  $z_1 + z_2$

Sol.  $z_1 + z_2 = 9 \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) + 5 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$

$$z_1 + z_2 = 9 \left[ -\frac{\sqrt{2}}{2} + i \left( \frac{-\sqrt{2}}{2} \right) \right] + 5 \left[ \frac{1}{2} + i \left( \frac{\sqrt{3}}{2} \right) \right]$$

$$z_1 + z_2 = 9 \left( -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \right) + 5 \left( \frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

$$z_1 + z_2 = -\frac{9\sqrt{2}}{2} - \frac{9\sqrt{2}}{2}i + \frac{5}{2} + \frac{5\sqrt{3}}{2}i$$

$$z_1 + z_2 = \left( -\frac{9\sqrt{2}}{2} + \frac{5}{2} \right) - \left( \frac{9\sqrt{2}}{2} - \frac{5\sqrt{3}}{2} \right)i$$

$$z_1 + z_2 = \frac{-9\sqrt{2} + 5}{2} - \frac{9\sqrt{2} - 5\sqrt{3}}{2}i$$

$$z_1 + z_2 = \frac{5 - 9\sqrt{2}}{2} - \frac{9\sqrt{2} - 5\sqrt{3}}{2}i$$

$$z_1 + z_2 = -3.86 - 2.03i$$

(ii)  $z_1 - z_2$

Sol.  $z_1 - z_2 = 9 \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) - 5 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$

$$z_1 - z_2 = 9 \left[ -\frac{\sqrt{2}}{2} + i \left( \frac{-\sqrt{2}}{2} \right) \right] - 5 \left[ \frac{1}{2} + i \left( \frac{\sqrt{3}}{2} \right) \right]$$

$$z_1 - z_2 = 9 \left( -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \right) - 5 \left( \frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

$$z_1 - z_2 = -\frac{9\sqrt{2}}{2} - \frac{9\sqrt{2}}{2}i - \frac{5}{2} - \frac{5\sqrt{3}}{2}i$$

$$z_1 - z_2 = -\frac{9\sqrt{2}}{2} - \frac{5}{2} - \frac{9\sqrt{2}}{2}i - \frac{5\sqrt{3}}{2}i$$

$$z_1 - z_2 = \left( -\frac{9\sqrt{2}}{2} - \frac{5}{2} \right) - \left( \frac{9\sqrt{2}}{2} + \frac{5\sqrt{3}}{2} \right)i$$

$$z_1 - z_2 = \frac{-9\sqrt{2} - 5}{2} - \frac{9\sqrt{2} + 5\sqrt{3}}{2}i$$

$$z_1 - z_2 = \frac{-5 - 9\sqrt{2}}{2} - \frac{9\sqrt{2} + 5\sqrt{3}}{2}i$$

$$z_1 - z_2 = -8.86 - 10.69i$$

(iii)  $z_1 \cdot z_2$

Sol.  $z_1 \cdot z_2 = 9 \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \cdot 5 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$

$$z_1 \cdot z_2 = 45 \left[ \cos \left( \frac{5\pi}{4} + \frac{\pi}{3} \right) + i \sin \left( \frac{5\pi}{4} + \frac{\pi}{3} \right) \right]$$

$$z_1 \cdot z_2 = 45 \left[ \cos \left( \frac{15\pi + 4\pi}{12} \right) + i \sin \left( \frac{15\pi + 4\pi}{12} \right) \right]$$

$$z_1 \cdot z_2 = 45 \left( \cos \frac{19\pi}{12} + i \sin \frac{19\pi}{12} \right)$$



$$(iv) \frac{z_1}{z_2}$$

$$\text{Sol. } \frac{z_1}{z_2} = \frac{9 \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right)}{5 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)}$$

$$\frac{z_1}{z_2} = \frac{9}{5} \left[ \cos \left( \frac{5\pi}{4} - \frac{\pi}{3} \right) + i \sin \left( \frac{5\pi}{4} - \frac{\pi}{3} \right) \right]$$

$$\frac{z_1}{z_2} = \frac{9}{5} \left[ \cos \left( \frac{15\pi - 4\pi}{12} \right) + i \sin \left( \frac{15\pi - 4\pi}{12} \right) \right]$$

$$\frac{z_1}{z_2} = \frac{9}{5} \left( \cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right)$$

**Q5.** If  $z_1 = 7 \left( \cos \frac{23\pi}{12} + i \sin \frac{23\pi}{12} \right)$  and

$$z_2 = 11 \left( \cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \text{ then find the}$$

following and express the result into  $x+iy$  form

(i)  $z_1 + z_2$

$$\text{Sol. } z_1 + z_2 = 7 \left( \cos \frac{23\pi}{12} + i \sin \frac{23\pi}{12} \right) + 11 \left( \cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right)$$

$$z_1 + z_2 = 7[(0.97) + i(-0.26)] + 11[(-0.97) + i(0.26)]$$

$$z_1 + z_2 = 7[(0.97 - 0.26i) + 11(-0.97 + 0.26i)]$$

$$z_1 + z_2 = 6.76 - 1.81i - 10.63 + 2.85i$$

$$z_1 + z_2 = (6.76 - 10.63) - (1.81 - 2.85)i$$

$$z_1 + z_2 = (-3.87) - (-1.04)i$$

$$z_1 + z_2 = -3.87 + 1.04i$$

(ii)  $z_1 - z_2$

$$\text{Sol. } z_1 - z_2 = 7 \left( \cos \frac{23\pi}{12} + i \sin \frac{23\pi}{12} \right) - 11 \left( \cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right)$$

$$z_1 - z_2 = 7[(0.97) + i(-0.26)] - 11[(-0.97) + i(0.26)]$$

$$z_1 - z_2 = 7[(0.97 - 0.26i) - 11(-0.97 + 0.26i)]$$

$$z_1 - z_2 = 6.76 - 1.81i + 10.63 - 2.85i$$

$$z_1 - z_2 = (6.76 + 10.63) - (1.81 + 2.85)i$$

$$z_1 - z_2 = (17.39) - (4.66)i$$

$$z_1 - z_2 = 17.39 - 4.66i$$

(iii)  $z_1 \cdot z_2$

$$\text{Sol. } z_1 \cdot z_2 = 7 \left( \cos \frac{23\pi}{12} + i \sin \frac{23\pi}{12} \right) \cdot 11 \left( \cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right)$$

$$z_1 \cdot z_2 = 77 \left[ \cos \left( \frac{23\pi}{12} + \frac{11\pi}{12} \right) + i \sin \left( \frac{23\pi}{12} + \frac{11\pi}{12} \right) \right]$$

$$z_1 \cdot z_2 = 77 \left( \cos \frac{23\pi + 11\pi}{12} + i \sin \frac{23\pi + 11\pi}{12} \right)$$

$$z_1 \cdot z_2 = 77 \left( \cos \frac{34\pi}{12} + i \sin \frac{34\pi}{12} \right)$$

$$z_1 \cdot z_2 = 77 \left( \cos \frac{17\pi}{6} + i \sin \frac{17\pi}{6} \right)$$

$$z_1 \cdot z_2 = 77 \left( \frac{-\sqrt{3}}{2} + i \frac{1}{2} \right)$$

$$z_1 \cdot z_2 = -\frac{77\sqrt{3}}{2} + \frac{77}{2}i$$

$$z_1 \cdot z_2 = -66.68 + 38.5i$$

(iv)  $\frac{z_1}{z_2}$

$$\text{Sol. } \frac{z_1}{z_2} = \frac{7 \left( \cos \frac{23\pi}{12} + i \sin \frac{23\pi}{12} \right)}{11 \left( \cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right)}$$

$$\frac{z_1}{z_2} = \frac{7}{11} \left[ \cos \left( \frac{23\pi}{12} - \frac{11\pi}{12} \right) + i \sin \left( \frac{23\pi}{12} - \frac{11\pi}{12} \right) \right]$$

$$\frac{z_1}{z_2} = \frac{7}{11} \left( \cos \frac{23\pi - 11\pi}{12} + i \sin \frac{23\pi - 11\pi}{12} \right)$$

$$\frac{z_1}{z_2} = \frac{7}{11} \left( \cos \frac{12\pi}{12} + i \sin \frac{12\pi}{12} \right)$$

$$\frac{z_1}{z_2} = \frac{7}{11} (\cos \pi + i \sin \pi)$$

$$\frac{z_1}{z_2} = \frac{7}{11} [(-1) + i(0)]$$

$$\frac{z_1}{z_2} = \frac{7}{11} [-1 + 0i]$$

$$\frac{z_1}{z_2} = -\frac{7}{11} + 0i$$

**Q6.** If  $z_1$  and  $z_2$  are two complex numbers, show that

(i)  $\text{Arg}(z_1 z_2) = \text{Arg } z_1 + \text{Arg } z_2$

**Sol.** L.H.S =  $\text{Arg}(z_1 z_2)$

$$\text{L.H.S} = \text{Arg} [(x_1 + y_1 i)(x_2 + y_2 i)]$$

$$\text{L.H.S} = \text{Arg} [(x_1 x_2 + x_1 y_2 i + x_2 y_1 i + y_1 y_2 i^2)]$$

$$\text{L.H.S} = \text{Arg} [x_1 x_2 + x_1 y_2 i + x_2 y_1 i - y_1 y_2]$$

$$\text{L.H.S} = \text{Arg} [x_1 x_2 - y_1 y_2 + i(x_1 y_2 + x_2 y_1)]$$

$$\text{L.H.S} = \tan^{-1} \left( \frac{x_1 y_2 + x_2 y_1}{x_1 x_2 - y_1 y_2} \right)$$

$$\text{R.H.S} = \text{Arg } z_1 + \text{Arg } z_2$$

$$\text{R.H.S} = \tan^{-1} \left( \frac{y_1}{x_1} \right) + \tan^{-1} \left( \frac{y_2}{x_2} \right)$$



$$R.H.S = \tan^{-1} \left( \frac{\frac{y_1}{x_1} + \frac{y_2}{x_2}}{1 - \left(\frac{y_1}{x_1}\right)\left(\frac{y_2}{x_2}\right)} \right)$$

$$\therefore \tan^{-1} A + \tan^{-1} B = \tan^{-1} \left( \frac{A+B}{1-AB} \right)$$

$$R.H.S = \tan^{-1} \left( \frac{\frac{y_1 x_2 + y_2 x_1}{x_1 x_2}}{1 - \frac{y_1 y_2}{x_1 x_2}} \right)$$

$$R.H.S = \tan^{-1} \left( \frac{\frac{y_1 x_2 + y_2 x_1}{x_1 x_2}}{\frac{x_1 x_2 - y_1 y_2}{x_1 x_2}} \right)$$

$$R.H.S = \tan^{-1} \left( \frac{y_1 x_2 + y_2 x_1}{x_1 x_2 - y_1 y_2} \right)$$

From (i) and (ii)

$$L.H.S = R.H.S$$

$$(ii) \quad \text{Arg} \left( \frac{z_1}{z_2} \right) = \text{Arg } z_1 - \text{Arg } z_2$$

Sol.  $L.H.S = \text{Arg} \left( \frac{z_1}{z_2} \right)$

$$L.H.S = \text{Arg} \left[ \frac{x_1 + y_1 i}{x_2 + y_2 i} \right]$$

$$L.H.S = \text{Arg} \left[ \frac{x_1 + y_1 i}{x_2 + y_2 i} \times \frac{x_2 - y_2 i}{x_2 - y_2 i} \right]$$

$$L.H.S = \text{Arg} \left[ \frac{x_1 x_2 - x_1 y_2 i + x_2 y_1 i - y_1 y_2 i^2}{(x_2)^2 - (y_2 i)^2} \right]$$

$$L.H.S = \text{Arg} \left[ \frac{x_1 x_2 - x_1 y_2 i + x_2 y_1 i + y_1 y_2}{x_2^2 + y_2^2} \right]$$

$$L.H.S = \text{Arg} \left[ \frac{x_1 x_2 + y_1 y_2 + i(x_2 y_1 - x_1 y_2)}{x_2^2 + y_2^2} \right]$$

$$L.H.S = \text{Arg} \left[ \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + \frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2} i \right]$$

$$L.H.S = \tan^{-1} \left( \frac{\frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2}}{\frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2}} \right)$$

$$L.H.S = \tan^{-1} \left( \frac{x_2 y_1 - x_1 y_2}{x_1 x_2 + y_1 y_2} \right) \quad (i)$$

$$R.H.S = \text{Arg } z_1 - \text{Arg } z_2$$

$$R.H.S = \tan^{-1} \left( \frac{y_1}{x_1} \right) - \tan^{-1} \left( \frac{y_2}{x_2} \right)$$

$$R.H.S = \tan^{-1} \left( \frac{\frac{y_1}{x_1} - \frac{y_2}{x_2}}{1 + \left(\frac{y_1}{x_1}\right)\left(\frac{y_2}{x_2}\right)} \right)$$

$$\therefore \tan^{-1} A - \tan^{-1} B = \tan^{-1} \left( \frac{A-B}{1+AB} \right)$$

$$R.H.S = \tan^{-1} \left( \frac{\frac{y_1 x_2 - y_2 x_1}{x_1 x_2}}{1 + \frac{y_1 y_2}{x_1 x_2}} \right)$$

$$R.H.S = \tan^{-1} \left( \frac{\frac{y_1 x_2 - y_2 x_1}{x_1 x_2}}{\frac{x_1 x_2 + y_1 y_2}{x_1 x_2}} \right)$$

$$R.H.S = \tan^{-1} \left( \frac{y_1 x_2 - y_2 x_1}{x_1 x_2 + y_1 y_2} \right) \quad (ii)$$

From (i) and (ii)

$$L.H.S = R.H.S$$

**Q7.** Divide  $z_1 = 6(\cos 150^\circ + i \sin 150^\circ)$  by  $z_2 = 3(\cos 30^\circ + i \sin 30^\circ)$  and express in  $x + iy$  form.

Sol.  $\frac{z_1}{z_2} = \frac{6(\cos 150^\circ + i \sin 150^\circ)}{3(\cos 30^\circ + i \sin 30^\circ)}$

$$\frac{z_1}{z_2} = \frac{6}{3} [\cos(150^\circ - 30^\circ) + i \sin(150^\circ - 30^\circ)]$$

$$\frac{z_1}{z_2} = 2 [\cos 120^\circ + i \sin 120^\circ]$$

$$\frac{z_1}{z_2} = 2 \left( -\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)$$

$$\frac{z_1}{z_2} = -1 + \sqrt{3}i$$

**Q8.** Multiply  $z_1 = 2(\cos 60^\circ + i \sin 60^\circ)$  and  $z_2 = 5(\cos 90^\circ + i \sin 90^\circ)$  and express in  $x + iy$  form.

Sol.  $z_1 z_2 = 2(\cos 60^\circ + i \sin 60^\circ) \cdot 5(\cos 90^\circ + i \sin 90^\circ)$

$$z_1 z_2 = 10 [\cos(60^\circ + 90^\circ) + i \sin(60^\circ + 90^\circ)]$$

$$z_1 z_2 = 10 [\cos 150^\circ + i \sin 150^\circ]$$

$$z_1 z_2 = 10 \left[ -\frac{\sqrt{3}}{2} + i \frac{1}{2} \right]$$

$$z_1 z_2 = -\frac{10\sqrt{3}}{2} + \frac{10}{2}i$$

$$z_1 z_2 = -5\sqrt{3} + 5i$$



**Q9.** Find the modulus and argument of  $z = -2 - 2i$ .

**Sol.** Here,  $x = -2, y = -2$

By using formula

$$|z| = \sqrt{x^2 + y^2}$$

$$|z| = \sqrt{(-2)^2 + (-2)^2}$$

$$|z| = \sqrt{4+4}$$

$$|z| = \sqrt{8}$$

$$|z| = 2\sqrt{2}$$

Also,  $\theta = \pi + \tan^{-1}\left(\frac{|y|}{|x|}\right) \because z$  lies in quadrant III

$$\theta = \pi + \tan^{-1}\left(\frac{|-2|}{|-2|}\right)$$

$$\theta = \pi + \tan^{-1}\left(\frac{2}{2}\right)$$

$$\theta = \pi + \tan^{-1}(1)$$

$$\theta = \pi + \frac{\pi}{4}$$

$$\theta = \frac{4\pi + \pi}{4}$$

$$\theta = \frac{5\pi}{4}$$

but the general argument is  $\frac{5\pi}{4} + 2n\pi$ , where  $n \in \mathbb{Z}$

Hence, the modulus of the given complex number is

$2\sqrt{2}$  and argument is  $\frac{5\pi}{4} + 2n\pi$ .

**Q10.** Write the equation  $\arg(\bar{z} - 2 + i) = \frac{2\pi}{3}$  in cartesian form, if  $z = x + iy$ .

**Sol.** Given,  $z = x + yi$

$$\Rightarrow \bar{z} = \overline{x + yi} = x - yi$$

Now, find the value of  $\bar{z} - 2 + i$

$$\bar{z} - 2 + i = (x - yi) - 2 + i$$

$$\bar{z} - 2 + i = x - yi - 2 + i$$

$$\bar{z} - 2 + i = x - 2 + i - yi$$

$$\bar{z} - 2 + i = (x - 2) + (1 - y)i$$

By using formula

$$\arg(z) = \tan^{-1}\left(\frac{b}{a}\right)$$

$\because a$  and  $b$  are real and imaginary parts of  $z$

$$\arg(\bar{z} - 2 + i) = \tan^{-1}\left(\frac{1-y}{x-2}\right)$$

$$\frac{2\pi}{3} = \tan^{-1}\left(\frac{1-y}{x-2}\right) \because \arg(\bar{z} - 2 + i) = \frac{2\pi}{3}$$

$$\tan \frac{2\pi}{3} = \frac{1-y}{x-2}$$

$$-\sqrt{3} = \frac{1-y}{x-2}$$

$$-\sqrt{3}(x-2) = 1-y$$

$$-\sqrt{3}x + 2\sqrt{3} = 1-y$$

$$y - \sqrt{3}x + 2\sqrt{3} - 1 = 0$$

$$y = \sqrt{3}x - 2\sqrt{3} + 1$$

**Q11.** If  $z = x + iy$  and  $\arg\left(\frac{\bar{z} - 1 + 2i}{z + 1 - 2i}\right) = \frac{9\pi}{4}$ , show that  $x^2 + y^2 - 4x + 2y - 5 = 0$ .

**Sol.**  $\arg\left(\frac{\bar{z} - 1 + 2i}{z + 1 - 2i}\right) = \frac{9\pi}{4}$

$$\arg\left(\frac{x + yi - 1 + 2i}{x + yi + 1 - 2i}\right) = \frac{9\pi}{4}$$

$$\arg\left(\frac{x - yi - 1 + 2i}{x - yi + 1 - 2i}\right) = \frac{9\pi}{4}$$

$$\arg\left[\frac{(x-1) + (2-y)i}{(x+1) - (2+y)i}\right] = \frac{9\pi}{4}$$

$$\arg\left[\frac{(x-1) + (2-y)i}{(x+1) - (2+y)i} \times \frac{(x+1) + (2+y)i}{(x+1) + (2+y)i}\right] = \frac{9\pi}{4}$$

$$\arg\left[\frac{(x-1)[(x+1) + (2+y)i] + (2-y)i[(x+1) + (2+y)i]}{(x+1)^2 - (2+y)^2 i^2}\right] = \frac{9\pi}{4}$$

$$\arg\left[\frac{(x^2-1) + (2x+xy-2-y)i + (2x+2-xy-y)i + (4-y^2)i^2}{(x^2+2x+1) - (4+4y+y^2)(-1)}\right] = \frac{9\pi}{4}$$

$$\arg\left[\frac{(x^2-1) + (2x+xy-2-y)i + (2x+2-xy-y)i + (4-y^2)(-1)}{(x^2+2x+1) + (4+4y+y^2)}\right] = \frac{9\pi}{4}$$

$$\arg\left[\frac{(x^2-1) + (2x+xy-2-y)i + (2x+2-xy-y)i - (4-y^2)}{(x^2+2x+1) + (4+4y+y^2)}\right] = \frac{9\pi}{4}$$

$$\arg\left[\frac{(x^2-1-4+y^2) + (2x+xy-2-y+2x+2-xy-y)i}{x^2+2x+1+4+4y+y^2}\right] = \frac{9\pi}{4}$$

$$\arg\left[\frac{(x^2-5+y^2) + (4x-2y)i}{x^2+2x+5+4y+y^2}\right] = \frac{9\pi}{4}$$

$$\arg\left(\frac{x^2-5+y^2}{x^2+2x+5+4y+y^2} + \frac{4x-2y}{x^2+2x+5+4y+y^2}i\right) = \frac{9\pi}{4}$$

$$\tan^{-1}\left(\frac{\frac{4x-2y}{x^2+2x+5+4y+y^2}}{\frac{x^2-5+y^2}{x^2+2x+5+4y+y^2}}\right) = \frac{9\pi}{4} \because \arg(z) = \tan^{-1}\left(\frac{b}{a}\right)$$



$$\frac{4x-2y}{x^2+2x+5+4y+y^2} = \tan \frac{9\pi}{4}$$

$$\frac{4x-2y}{x^2-5+y^2} = 1$$

$$4x-2y = x^2-5+y^2$$

$$0 = x^2+y^2-4x+2y-5$$

Hence, proved.

**Q12.** If  $z = x + iy$  and  $\arg(z-2-3i) - \arg(z+2+3i) = 2\pi$ , show that  $2y = 3x$ .

**Sol.**  $\arg(z-2-3i) - \arg(z+2+3i) = 2\pi$   
 $\arg(x+yi-2-3i) - \arg(x+yi+2+3i) = 2\pi$   
 $\arg[(x-2) + (y-3)i] - \arg[(x+2) + (y+3)i] = 2\pi$   
 $\tan^{-1}\left(\frac{y-3}{x-2}\right) - \tan^{-1}\left(\frac{y+3}{x+2}\right) = 2\pi \therefore \arg(z) = \tan^{-1}\left(\frac{y}{x}\right)$

$$\tan^{-1}\left[\frac{\frac{y-3}{x-2} - \frac{y+3}{x+2}}{1 + \left(\frac{y-3}{x-2}\right)\left(\frac{y+3}{x+2}\right)}\right] = 2\pi$$

$$\therefore \tan^{-1} A - \tan^{-1} B = \tan^{-1}\left(\frac{A-B}{1+AB}\right)$$

$$\tan^{-1}\left[\frac{\frac{(y-3)(x+2) - (y+3)(x-2)}{x^2-4}}{1 + \frac{y^2-9}{x^2-4}}\right] = 2\pi$$

$$\tan^{-1}\left[\frac{\frac{xy+2y-3x-6 - (xy-2y+3x-6)}{x^2-4}}{\frac{x^2-4+y^2-9}{x^2-4}}\right] = 2\pi$$

$$\tan^{-1}\left(\frac{xy+2y-3x-6 - xy+2y-3x+6}{x^2-4+y^2-9}\right) = 2\pi$$

$$\tan^{-1}\left(\frac{4y-6x}{x^2+y^2-13}\right) = 2\pi$$

$$\frac{4y-6x}{x^2+y^2-13} = \tan 2\pi$$

$$\frac{4y-6x}{x^2+y^2-13} = 0$$

$$4y-6x = 0$$

$$2(2y-3x) = 0$$

$$2y-3x = 0$$

$$2y = 3x$$

Hence, proved.

**Q13.** Solve the equation  $|z-2i| = |\bar{z}+2|$  for  $z = x + iy$ .

**Sol.**  $|x+yi-2i| = |\overline{x+yi}+2|$   
 $|x+yi-2i| = |x-yi+2|$   
 $|x+(y-2)i| = |(x+2)-yi|$   
 $\sqrt{x^2+(y-2)^2} = \sqrt{(x+2)^2+(-y)^2}$   
 $\sqrt{x^2+y^2-4y+4} = \sqrt{x^2+4x+4+y^2}$   
 $\sqrt{x^2+y^2-4y+4} = \sqrt{x^2+y^2+4x+4}$   
 $(\sqrt{x^2+y^2-4y+4})^2 = (\sqrt{x^2+y^2+4x+4})^2$   
 $x^2+y^2-4y+4 = x^2+y^2+4x+4$   
 $x^2+y^2-4y+4 - x^2-y^2-4x-4 = 0$   
 $-4y-4x = 0$   
 $-4(y+x) = 0$   
 $y+x = 0 \Rightarrow y = -x$

**Q14.** For  $z = x + iy$ , solve the equation  $|5z+4+i| = |5\bar{z}-3+2i|$ .

**Sol.**  $|5(x+yi)+4+i| = |5(\overline{x+yi})-3+2i|$   
 $|5x+5yi+4+i| = |5\bar{x}-5\bar{y}i-3+2i|$   
 $|5x+4 + (5y+1)i| = |5\bar{x}-5\bar{y}i-3+2i|$   
 $|(5x+4) + (5y+1)i| = |(5\bar{x}-3) + (2-5\bar{y})i|$   
 $\sqrt{(5x+4)^2 + (5y+1)^2} = \sqrt{(5\bar{x}-3)^2 + (2-5\bar{y})^2}$   
 $\sqrt{25x^2+40x+16+25y^2+10y+1} = \sqrt{25\bar{x}^2-30\bar{x}+9+4-20\bar{y}+25\bar{y}^2}$   
 $\sqrt{25x^2+25y^2+40x+10y+17} = \sqrt{25x^2+25y^2-30x-20y+13}$   
 $(\sqrt{25x^2+25y^2+40x+10y+17})^2 = (\sqrt{25x^2+25y^2-30x-20y+13})^2$   
 $25x^2+25y^2+40x+10y+17 = 25x^2+25y^2-30x-20y+13$   
 $25x^2+25y^2+40x+10y+17 - 25x^2-25y^2+30x+20y-13 = 0$   
 $70x+30y+4 = 0$   
 $2(35x+15y+2) = 0$   
 $35x+15y+2 = 0$   
 $15y = -35x-2 \Rightarrow y = \frac{-35x-2}{15}$

**Q15.** Determine the set of points  $z = x + iy$  that satisfy the equation  $|3\bar{z}-2+i| = |3z+i|$ .

**Sol.**  $|3\bar{z}-2+i| = |3z+i|$   
 $|3(\overline{x+yi})-2+i| = |3(x+yi)+i|$   
 $|3(x-yi)-2+i| = |3x+3yi+i|$   
 $|3x-3yi-2+i| = |3x+(3y+1)i|$   
 $|(3x-2) + (1-3y)i| = |3x+(3y+1)i|$   
 $\sqrt{(3x-2)^2 + (1-3y)^2} = \sqrt{(3x)^2 + (3y+1)^2}$   
 $\sqrt{9x^2-12x+4+1-6y+9y^2} = \sqrt{9x^2+9y^2+6y+1}$



$$\sqrt{9x^2 - 12x + 5 - 6y + 9y^2} = \sqrt{9x^2 + 9y^2 + 6y + 1}$$

$$(\sqrt{9x^2 - 12x + 5 - 6y + 9y^2})^2 = (\sqrt{9x^2 + 9y^2 + 6y + 1})^2$$

$$9x^2 - 12x + 5 - 6y + 9y^2 = 9x^2 + 9y^2 + 6y + 1$$

$$9x^2 - 12x + 5 - 6y + 9y^2 - 9x^2 - 9y^2 - 6y - 1 = 0$$

$$-12x + 4 - 12y = 0 \Rightarrow -4(3x - 1 + 3y) = 0$$

$$3x - 1 + 3y = 0 \Rightarrow 3x = -3y + 1$$

$$y = \frac{-3x+1}{3} \Rightarrow y = -\frac{3x}{3} + \frac{1}{3} \Rightarrow y = -x + \frac{1}{3}$$

**Q16.** If  $z = x + iy$  and  $w = \frac{1-iz}{z-i}$ , show that

$$|w| = 1 \Rightarrow z \text{ is real.}$$

Sol. As, given  $|w| = 1$

$$\left| \frac{1-iz}{z-i} \right| = 1$$

$$\left| \frac{1-i(x+yi)}{(x+yi)-i} \right| = 1$$

$$\left| \frac{1-xi-yi^2}{x+yi-i} \right| = 1$$

$$\left| \frac{1-xi+y}{x+yi-i} \right| = 1$$

$$\left| \frac{1+y-xi}{x+i(y-1)} \right| = 1$$

$$\left| \frac{1+y-xi}{x+i(y-1)} \right| = 1$$

$$\frac{\sqrt{(1+y)^2 + (-x)^2}}{\sqrt{(x)^2 + (y-1)^2}} = 1$$

$$\frac{\sqrt{1+y^2+2y+x^2}}{\sqrt{x^2+1+y^2-2y}} = 1$$

$$\sqrt{1+y^2+2y+x^2} = \sqrt{x^2+1+y^2-2y}$$

By taking square on both sides

$$(\sqrt{1+y^2+2y+x^2})^2 = (\sqrt{x^2+1+y^2-2y})^2$$

$$1+y^2+2y+x^2 = x^2+1+y^2-2y$$

$$1+y^2+2y+x^2-x^2-1-y^2+2y=0$$

$$4y=0 \Rightarrow y=0$$

Now put the value of  $y$  in  $z = x + yi$

$$z = x + 0i$$

$$z = x$$

Which shows that  $z$  is a real number.

**Q17.** If  $z_1$  and  $z_2$  are different complex

numbers with  $|z_2| = 1$ , find  $\left| \frac{z_2 - z_1}{1 - \overline{z_1} z_2} \right|$

$$\text{Sol.} = \left| \frac{z_2 - z_1}{1 - \overline{z_1} z_2} \right| = \left| \frac{z_2 - z_1}{1 - \overline{z_1} z_2} \right|$$

$$= \frac{|z_2 - z_1|}{|1 - \overline{z_1} z_2|} \quad \because |z| = |\overline{z}|$$

$$= \frac{|z_2 - z_1|}{|1 - \overline{z_1} z_2|} \quad \because \overline{z - w} = \overline{z} - \overline{w}$$

$$= \frac{|z_2 - z_1|}{|1 - \overline{z_1} z_2|} \quad \because \overline{zw} = \overline{z} \overline{w}$$

$$= \frac{|z_2 - z_1|}{|1 - \overline{z_1} z_2|} \quad \because \overline{\overline{z}} = z$$

$$= \frac{|z_2 - z_1|}{|1 - \overline{z_1} \frac{\overline{z_2}}{|z_2|}|} \quad \because \frac{1}{z} = \frac{\overline{z}}{|z|} \Rightarrow \frac{|z|}{z} = \overline{z}$$

$$= \frac{|z_2 - z_1|}{|1 - \frac{\overline{z_1}}{z_2}|} \quad \because |z_2| = 1$$

$$= \frac{|z_2 - z_1|}{\frac{|z_2 - z_1|}{|z_2|}} = \frac{|z_2 - z_1|}{|z_2|}$$

$$= \frac{|z_2 - z_1|}{|z_2 - z_1|} \quad \because |z_2| = 1 \Rightarrow = 1$$

**Q18.** An AC source supplies a voltage of

$$V = 120 \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \text{ volts to a circuit with}$$

impedance  $Z = \frac{1+i\sqrt{3}}{2}$  ohms. Calculate the current in polar form.

$$\text{Sol. Given } Z = \frac{1+i\sqrt{3}}{2}$$

$$Z = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$\text{Here, } x = \frac{1}{2}, y = \frac{\sqrt{3}}{2}$$

By using the formula

$$r = \sqrt{x^2 + y^2}$$



$$r = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$r = \sqrt{\frac{1}{4} + \frac{3}{4}}$$

$$r = \sqrt{\frac{1+3}{4}}$$

$$r = \sqrt{\frac{4}{4}}$$

$$r = \sqrt{1}$$

$$r = 1$$

$$\text{Also, } \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\theta = \tan^{-1}\left(\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}\right)$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right)$$

$$\theta = \frac{\pi}{3}$$

Now,  $x + yi = r(\cos \theta + i \sin \theta)$

$$\frac{1}{2} + \frac{\sqrt{3}}{2}i = 1\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$$

$$Z = 1\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$$

By using the formula

$$V = IZ$$

$$120\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right) = I\left[1\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)\right]$$

$$\frac{120\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)}{1\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)} = I$$

$$120\left[\cos\left(\frac{\pi}{4} - \frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{4} - \frac{\pi}{3}\right)\right] = I$$

$$120\left(\cos \frac{3\pi - 4\pi}{12} + i \sin \frac{3\pi - 4\pi}{12}\right) = I$$

$$120\left(\cos \frac{-\pi}{12} + i \sin \frac{-\pi}{12}\right) = I$$

$$120\left(\cos \frac{\pi}{12} - i \sin \frac{\pi}{12}\right) = I$$

**019.** An AC circuit has an impedance of  $Z = 3 - 6i$  ohms and is connected to a voltage source of  $V = 90 + 30i$  volts. Find the current in both rectangular and polar form.

**Sol.** By using the formula

$$V = IZ$$

$$90 + 30i = I(3 - 6i)$$

$$\frac{90 + 30i}{3 - 6i} = I$$

$$\frac{90 + 30i}{3 - 6i} \times \frac{3 + 6i}{3 + 6i} = I$$

$$\frac{270 + 540i + 90i + 180i^2}{3^2 - (6i)^2} = I$$

$$\frac{270 + 540i + 90i + 180(-1)}{9 - 36i^2} = I$$

$$\frac{90 + 630i}{9 + 36} = I$$

$$\frac{90 + 630i}{45} = I$$

$$\frac{90}{45} + \frac{630}{45}i = I$$

$$2 + 14i = I$$

Here,  $x = 2$ ,  $y = 14$

By using formula

$$r = \sqrt{x^2 + y^2}$$

$$r = \sqrt{2^2 + 14^2}$$

$$r = \sqrt{4 + 196}$$

$$r = \sqrt{200}$$

$$r = 10\sqrt{2}$$

$$\text{Also, } \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\theta = \tan^{-1}\left(\frac{14}{2}\right)$$

$$\theta = \tan^{-1}(7)$$

$$\theta = 81.87^\circ$$

Now,  $x + yi = r(\cos \theta + i \sin \theta)$

$$2 + 14i = 10\sqrt{2}(\cos 81.87^\circ + i \sin 81.87^\circ)$$

$$I = 10\sqrt{2}(\cos 81.87^\circ + i \sin 81.87^\circ)$$



**Q20.** Encrypt the word "CODE" by multiplying the complex encryption key  $k = 2 - i$ . Then decrypt it back to the original word.

**Sol.** First, we assign each letter in the word "CODE" a complex number with zero imaginary part. The encryption and decryption shown in the table below.

| Letter | Complex Number $z$ | $z$ Encrypted $= z \times k$                                   | $z$ Decrypted $= z \text{ Encrypted} \div k$                                                                                                                                                                                                                                                                                                                       | Letter |
|--------|--------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| C      | $3 + 0i$           | $(3 + 0i)(2 - i) = (3)(2 - i)$                                 | $\frac{6 - 3i}{2 - i} = \frac{6 - 3i}{2 - i} \times \frac{2 + i}{2 + i}$ $\frac{6 - 3i}{2 - i} = \frac{12 + 6i - 6i - 3i^2}{2^2 - i^2}$ $\frac{6 - 3i}{2 - i} = \frac{12 - 3(-1)}{4 - (-1)}$ $\frac{6 - 3i}{2 - i} = \frac{12 + 3}{4 + 1}$ $\frac{6 - 3i}{2 - i} = \frac{15}{5}$ $\frac{6 - 3i}{2 - i} = 3$ $\frac{6 - 3i}{2 - i} = 3 + 0i$                        | C      |
| O      | $15 + 0i$          | $(15 + 0i)(2 - i) = (15)(2 - i)$ $(15 + 0i)(2 - i) = 30 - 15i$ | $\frac{30 - 15i}{2 - i} = \frac{30 - 15i}{2 - i} \times \frac{2 + i}{2 + i}$ $\frac{30 - 15i}{2 - i} = \frac{60 + 30i - 30i - 15i^2}{2^2 - i^2}$ $\frac{30 - 15i}{2 - i} = \frac{60 - 15(-1)}{4 - (-1)}$ $\frac{30 - 15i}{2 - i} = \frac{60 + 15}{4 + 1}$ $\frac{30 - 15i}{2 - i} = \frac{75}{5}$ $\frac{30 - 15i}{2 - i} = 15$ $\frac{30 - 15i}{2 - i} = 15 + 0i$ | O      |
| D      | $4 + 0i$           | $(4 + 0i)(2 - i) = (4)(2 - i)$ $(4 + 0i)(2 - i) = 8 - 4i$      | $\frac{8 - 4i}{2 - i} = \frac{8 - 4i}{2 - i} \times \frac{2 + i}{2 + i}$ $\frac{8 - 4i}{2 - i} = \frac{16 + 8i - 8i - 4i^2}{2^2 - i^2}$ $\frac{8 - 4i}{2 - i} = \frac{16 - 4(-1)}{4 - (-1)}$ $\frac{8 - 4i}{2 - i} = \frac{16 + 4}{4 + 1}$ $\frac{8 - 4i}{2 - i} = \frac{20}{5}$                                                                                   | D      |



|   |        |                                                |                                                                                                                                                                                                                                                                                                 |   |
|---|--------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   |        |                                                | $\frac{8-4i}{2-i} = 4$ $\frac{8-4i}{2-i} = 4+0i$                                                                                                                                                                                                                                                |   |
| E | $5+0i$ | $(5+0i)(2-i) = (5)(2-i)$ $(5+0i)(2-i) = 10-5i$ | $\frac{10-5i}{2-i} = \frac{10-5i}{2-i} \times \frac{2+i}{2+i}$ $\frac{10-5i}{2-i} = \frac{20+10i-10i-5i^2}{2^2-i^2}$ $\frac{10-5i}{2-i} = \frac{20-5(-1)}{4-(-1)}$ $\frac{10-5i}{2-i} = \frac{20+5}{4+1}$ $\frac{10-5i}{2-i} = \frac{25}{5}$ $\frac{10-5i}{2-i} = 5$ $\frac{10-5i}{2-i} = 5+0i$ | E |

**021.** Consider the complex encryption key  $k = 3 - 3i$ . Encrypt the word "QUIZ", and then recover the original word using the inverse of the key.

**Sol.** First, we assign each letter in the word "QUIZ" a complex number with zero imaginary part. The encryption and decryption shown in the table below.

| Letter | Complex Number $z$ | $z$ Encrypted = $z \times k$                          | $z$ Decrypted = $z \text{ Encrypted} \div k$                                                                                                                                                                                                                                                                                           | Letter |
|--------|--------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Q      | $17+0i$            | $(17+0i)(3-3i) = (17)(3-3i)$ $(17+0i)(3-3i) = 51-51i$ | $\frac{51-51i}{3-3i} = \frac{51-51i}{3-3i} \times \frac{3+3i}{3+3i}$ $\frac{51-51i}{3-3i} = \frac{153+153i-153i-153i^2}{(3)^2-(3i)^2}$ $\frac{51-51i}{3-3i} = \frac{153-153(-1)}{9-9(-1)}$ $\frac{51-51i}{3-3i} = \frac{153+153}{9+9}$ $\frac{51-51i}{3-3i} = \frac{306}{18}$ $\frac{51-51i}{3-3i} = 17$ $\frac{51-51i}{3-3i} = 17+0i$ | Q      |
| U      | $21+0i$            | $(21+0i)(3-3i) = (21)(3-3i)$ $(21+0i)(3-3i) = 63-63i$ | $\frac{63-63i}{3-3i} = \frac{63-63i}{3-3i} \times \frac{3+3i}{3+3i}$ $\frac{63-63i}{3-3i} = \frac{189+189i-189i-189i^2}{(3)^2-(3i)^2}$ $\frac{63-63i}{3-3i} = \frac{189-189(-1)}{9-9(-1)}$                                                                                                                                             | U      |



|   |         |                                                       |                                                                                                                                                                                                                                                                                                                                        |   |
|---|---------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   |         |                                                       | $\frac{63-63i}{3-3i} = \frac{189+189}{9+9}$ $\frac{63-63i}{3-3i} = \frac{378}{18}$ $\frac{63-63i}{3-3i} = 21$ $\frac{63-63i}{3-3i} = 21+0i$                                                                                                                                                                                            |   |
| I | $9+0i$  | $(9+0i)(3-3i) = (9)(3-3i)$ $(9+0i)(3-3i) = 27-27i$    | $\frac{27-27i}{3-3i} = \frac{27-27i}{3-3i} \times \frac{3+3i}{3+3i}$ $\frac{27-27i}{3-3i} = \frac{81+81i-81i-81i^2}{(3)^2-(3i)^2}$ $\frac{27-27i}{3-3i} = \frac{81-81(-1)}{9-9(-1)}$ $\frac{27-27i}{3-3i} = \frac{81+81}{9+9}$ $\frac{27-27i}{3-3i} = \frac{162}{18}$ $\frac{27-27i}{3-3i} = 9$ $\frac{27-27i}{3-3i} = 9+0i$           | I |
| Z | $26+0i$ | $(26+0i)(3-3i) = (26)(3-3i)$ $(26+0i)(3-3i) = 78-78i$ | $\frac{78-78i}{3-3i} = \frac{78-78i}{3-3i} \times \frac{3+3i}{3+3i}$ $\frac{78-78i}{3-3i} = \frac{234+234i-234i-234i^2}{(3)^2-(3i)^2}$ $\frac{78-78i}{3-3i} = \frac{234-234(-1)}{9-9(-1)}$ $\frac{78-78i}{3-3i} = \frac{234+234}{9+9}$ $\frac{78-78i}{3-3i} = \frac{468}{18}$ $\frac{78-78i}{3-3i} = 26$ $\frac{78-78i}{3-3i} = 26+0i$ | Z |

**022.** Encrypt the word "CLASS" by adding the complex encryption key  $k = -3 + 4i$ . Then decrypt it back to the original word.

**Sol.** First, we assign each letter in the word "CLASS" a complex number with zero imaginary part. The encryption and decryption shown in the table below.

| Letter | Complex Number $z$ | $z$ Encrypted $= z + k$                     | $z$ Decrypted $= z \text{ Encrypted} - k$                   | Letter |
|--------|--------------------|---------------------------------------------|-------------------------------------------------------------|--------|
| C      | $3+0i$             | $3+0i+(-3+4i) = 3-3+4i$ $3+0i+(-3+4i) = 4i$ | $4i-(-3+4i) = 4i+3-4i$ $4i-(-3+4i) = 3$ $4i-(-3+4i) = 3+0i$ | C      |



|   |           |                                                                        |                                                                                                                                             |   |
|---|-----------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---|
| L | $12 + 0i$ | $12 + 0i + (-3 + 4i) = 12 - 3 + 4i$<br>$12 + 0i + (-3 + 4i) = 9 + 4i$  | $9 + 4i - (-3 + 4i) = 9 + 4i + 3 - 4i$<br>$9 + 4i - (-3 + 4i) = 9 + 3$<br>$9 + 4i - (-3 + 4i) = 12$<br>$9 + 4i - (-3 + 4i) = 12 + 0i$       | L |
| A | $1 + 0i$  | $1 + 0i + (-3 + 4i) = 1 - 3 + 4i$<br>$1 + 0i + (-3 + 4i) = -2 + 4i$    | $-2 + 4i - (-3 + 4i) = -2 + 4i + 3 - 4i$<br>$-2 + 4i - (-3 + 4i) = -2 + 3$<br>$-2 + 4i - (-3 + 4i) = 1$<br>$-2 + 4i - (-3 + 4i) = 1 + 0i$   | A |
| S | $19 + 0i$ | $19 + 0i + (-3 + 4i) = 19 - 3 + 4i$<br>$19 + 0i + (-3 + 4i) = 16 + 4i$ | $16 + 4i - (-3 + 4i) = 16 + 4i + 3 - 4i$<br>$16 + 4i - (-3 + 4i) = 16 + 3$<br>$16 + 4i - (-3 + 4i) = 19$<br>$16 + 4i - (-3 + 4i) = 19 + 0i$ | S |
| S | $19 + 0i$ | $19 + 0i + (-3 + 4i) = 19 - 3 + 4i$<br>$19 + 0i + (-3 + 4i) = 16 + 4i$ | $16 + 4i - (-3 + 4i) = 16 + 4i + 3 - 4i$<br>$16 + 4i - (-3 + 4i) = 16 + 3$<br>$16 + 4i - (-3 + 4i) = 19$<br>$16 + 4i - (-3 + 4i) = 19 + 0i$ | S |

### Important Formulas to Remember

- $(a + ib) + (c + id) = (a + c) + i(b + d)$
- $k(a + ib) = ka + ikb$
- $(a + ib) - (c + id) = (a + ib) + [-(c + id)] = a + ib + (-c - id) = (a - c) + i(b - d)$
- $(a + ib)(c + id) = ac + iad + ibc + i^2bd = (ac - bd) + i(ad + bc)$
- $|x + iy| = \sqrt{x^2 + y^2}$
- $\sqrt{x + yi} = \pm \left( \sqrt{\frac{\sqrt{x^2 + y^2} + x}{2}} + \frac{iy}{|y|} \sqrt{\frac{\sqrt{x^2 + y^2} - x}{2}} \right)$
- If  $\omega = \frac{-1 + \sqrt{3}i}{2}$ , then  $\omega^2 = \frac{-1 - \sqrt{3}i}{2}$
- $x + iy = r \cos \theta + i r \sin \theta$
- $\text{Arg } z = \theta = \tan^{-1} \left( \frac{b}{a} \right), (a \neq 0)$
- $z_1 + z_2 = r_1(\cos \theta_1 + i \sin \theta_1) + r_2(\cos \theta_2 + i \sin \theta_2)$
- $z_1 - z_2 = r_1(\cos \theta_1 + i \sin \theta_1) - r_2(\cos \theta_2 + i \sin \theta_2)$
- $z_1 \cdot z_2 = r_1 \cdot r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$
- $\arg(z_1 \cdot z_2) = \arg(z_1) + \arg(z_2)$
- $\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$
- $\arg \left( \frac{z_1}{z_2} \right) = \arg(z_1) - \arg(z_2)$

### Objective Type Questions

#### Multiple Choice Questions (MCQs)

#### 1.1 Complex Numbers

- The set of complex numbers is represented by:  
 (A)  $R$  (B)  $Q$  (C)  $Z$  (D)  $C$
- Every real number is a \_\_\_\_\_ number.  
 (A) Natural (B) Rational (C) Irrational (D) Complex
- If  $z = a + bi$  then which statement is true?  
 (A)  $\text{Re}(z) = a$  and  $\text{Im}(z) = bi$  (B)  $\text{Re}(z) = a$  and  $\text{Im}(z) = b$   
 (C)  $\text{Re}(z) = bi$  and  $\text{Im}(z) = a$  (D)  $\text{Re}(z) = b$  and  $\text{Im}(z) = a$